

Best Lightweight, No-GPU OCR Engines & Garble Correctors — Expanded Edition

Webtoon · Manhwa · Manga · Comic Speech Bubbles

Deep research including primary engines, hidden gems, pure-Rust/WASM ports, ultra-light models & post-OCR correctors

Compiled 2026-09-10 · CPU-only · Fast · Minimal garbles

Scope: Tools that run without a dedicated GPU, stay lightweight/fast, and minimize garbled text on stylized comic fonts, vertical text, speech bubbles, SFX and long-strip webtoon layouts (Korean manhwa, Chinese manhua, Japanese manga, English comics).

1. Why Generic OCR Fails on Comic / Webtoon Text

Tesseract, basic EasyOCR and cloud document engines are trained on clean horizontal print. Comic text breaks those assumptions: vertical (tategaki) and mixed-direction text; highly stylized/hand-drawn fonts; irregular speech bubbles over complex art; furigana/ruby and SFX; long vertical webtoon strips; color, gradients and screentones. Result: high CER and heavy garble. Specialized models + bubble cropping dramatically cut errors before any post-correction.

2. Primary Lightweight No-GPU Engines (Ranked)

Rank	Engine	Size / Deps	Langs	Comic Strength	CPU
1	Baberu OCR	115M · 121–242 MB ONNX onnxruntime+numpy+pillow	JA/ZH/EN	Bubble specialist; beats manga-ocr on JA	Very fast
2	Hayai OCR v2.1	Quantizable + LiteRT/TFLite (no PyTorch option)	JA/ZH/KR	Multi-line bubbles, vertical, furigana, SFX, KR	Fast
3	RapidOCR (PP-OCRv5)	~50–80 MB ONNX	Strong CJK + EN	Used in real manhwa pipelines	Extremely fast
4	manga-ocr-torchless	~400 MB ONNX (no Torch)	Japanese	Classic multi-line bubble OCR, Torch-free	Fast
5	XianScan pipeline	Self-contained ~1 GB binary	Multi-lang incl. KR	CPU-first detect+OCR+inpaint+typeset; long-strip	Smooth

2.1 Baberu OCR

115M model for manga speech bubbles. Lowest ICER on Manga109-v2026 among compared models; competitive with 0.9B VL on ZH/EN typical bubbles. Character-level 14 630 vocab keeps SFX/rare chars intact.

```
pip install onnxruntime numpy pillow
huggingface-cli download genshiai-daichi/baberu-ocr --local-dir baberu-ocr
python onnx_infer.py --image bubble.jpg # or --vision vision_int4.onnx for 121 MB
```

HF: [genshiai-daichi/baberu-ocr](#) · Apache-2.0

2.2 Hayai OCR

Asian-comics specialist (vertical/horizontal, furigana, SFX, multi-line). Strong CER on JMangaBench. LiteRT backend = pure CPU, no PyTorch.

```
pip install hayai-ocr[litert]
from hayai_ocr import HayaiOcr
mocr = HayaiOcr(backend="litert") # or litert_quant="dynamic_wi4"
```

GitHub: [NopeNopeGuy/hayai-ocr](#)

2.3 RapidOCR + XianScan + manga-ocr-torchless

RapidOCR: ONNX Paddle models, tiny footprint, primary in manhwa-recap pipelines. **XianScan:** Rust+ONNX self-contained binary, long-strip webtoon handling, Korean dict support, CPU-first (4-core AVX2 + 8 GB). **manga-ocr-torchless:** ONNX drop-in of classic manga-ocr with parity, no Torch.

3. Hidden Gems & Underrated Tools (Deep Research)

Lesser-known engines that fill niches the primaries miss (extreme size, pure Rust, Korean-only, browser/WASM, etc.).

Tool	Size	Why a gem	Best for
gnurt2041/MangaOCR	~8 MB	Tiny end-to-end JA manga model (vert/horiz, furigana, overlaid)	Edge / absolute minimal JA
manga-ocr-rs (Rust)	~441 MB models	Pure Rust, no Python; WASM-capable; horiz/vert/handwritten	Embeddable / no-Python JA
yurvon/japanese-ocr	ONNX Rust	Pure Rust WASM manga OCR port (ViT+BERT)	WASM / desktop JA
MIT 48px CTC ONNX	Small CTC	Actual recognizer from manga-image-translator, clean ONNX	Drop-in after any detector
trocr-small-korean	54.5 M	Small TrOCR fine-tuned for Korean	KR bubble crops
Korean Light-OCR (jientai)	~12 MB total	MobileNetV3 det+rec (~90% on their tests), H+V models	Extreme light Korean
chineseocr_lite	~4.7 MB total	Classic ultra-light CN OCR + vertical support (ONNX/NCNN/MNN)	Manhua / mobile CN
ComiQ	1.5–34 M tiers	Comic-focused hybrid OCR + structured schemas (speaker etc.)	Structured comic extraction
ocrs (Rust)	ONNX/RTen	Modern general OCR, easy WASM, open datasets	General + WASM
eSearch-OCR / OnnxOCR	ONNX packs	Paddle-derived pure ONNX, multi-lang packs incl. KR/JA	Framework-free Paddle accuracy
NDLOCR-Lite	CPU native	National Diet Library lightweight JA OCR, commercial OK (CC BY)	Offline JA documents / books
COMICS Text+ models	FCENet+MASTER	Fine-tuned detection+recognition on Western comics gold standard	English/Western comics research

Additional notes: **WORD-pytorch** (webtoon-specific bubble+line+KR recognizer, older but still usable); **ogkalu** YOLO/RT-DETR comic detectors (still under-used outside a few translators); **Kiuyha Manga-Bubble-YOLO** nano/small; **Untrusted-Translations-Android** ships Baberu + RapidOCR packs on-device.

4. Detection First — Highest Leverage Anti-Garble Step

Full-page OCR produces far more garbles than OCR on tightly cropped bubbles. Recommended detectors: ogkalu/comic-text-and-bubble-detector (RT-DETR), ogkalu YOLOv8 bubble/text segmenters, Koharu RF-DETR, YOLO26-Nano (ComicTL, 2.4 M params), CTD (Comic Text Detector). Crop → specialized OCR is the single highest-ROI technique.

Always crop to speech bubbles / text regions before recognition.

5. Lightweight Garble / Post-OCR Correctors

ZentrixLabs.OcrCorrection (.NET): ~865 curated English patterns (pipe→|, |↔|, spacing, apostrophes). Multi-pass, ~1 s for feature-length text, near-zero false positives. Ideal for English webtoon dialogue and subtitle-style OCR. **Macula** (Zig→WASM): client-side visual confusion rules (rn→m, cl→d, 0↔O). COMICS Text+ rule-based post-process (frequency vocab + first/last letter + single-char cleanup). Dictionary + edit-distance / Hunspell + comic slang lexicon remain the fastest pure-CPU options.

Zentrix: github.com/ZentrixLabs/ZentrixLabs.OcrCorrection

6. Recommended Stacks by Use Case

Use Case	Stack	Why
Pure light + bubbles	Crop → Baberu (ONNX) → optional Zentrix	Smallest accurate bubble reader
Korean manhwa	Hayai LiteRT or RapidOCR + XianScan long-strip	Native KR + vertical strip
Bulk mixed CJK+EN	RapidOCR PP-OCRv5 ONNX + detector	Fastest practical accuracy
Japanese classic	manga-ocr-torchless / Baberu / manga-ocr-rs	Proven multi-line quality
English residual	Any OCR + Zentrix or Macula	Pattern correctors shine
Absolute minimal size	gnurt MangaOCR (~8 MB) or chineseocr_lite (~4.7 MB)	Edge / low-RAM
No Python / embed	manga-ocr-rs or pure-Rust/japanese-ocr	Native binary / WASM
Browser / zero install	ComicTL (YOLO26 + PaddleOCR ONNX Web) or Yomikomi	On-device privacy
Full auto chapters	XianScan self-contained binary	Detect→OCR→inpaint→typeset

7. Practical Tips to Further Cut Garbles

- Always detect & crop bubbles before OCR.
- Prefer ONNX / LiteRT / quantized / MNN builds — keep accuracy, stay light.
- Light preprocessing on crops (contrast, mild skew) helps stylized fonts.
- Long webtoon strips: use gutter-aware tools (XianScan) or split panels first.
- Keep a domain dictionary of names, slang and SFX for post-correction.

- Avoid large VLMs if strict no-GPU / low memory is required; 50–250 MB specialists are the sweet spot.
- All recommended tools are offline after first model download.

8. Key Links (Quick Reference)

Baberu OCR: huggingface.co/genshiai-daichi/baberu-ocr

Hayai OCR: github.com/NopeNopeGuy/hayai-ocr

RapidOCR: github.com/RapidAI/RapidOCR

manga-ocr-torchless: github.com/liksunrice/manga-ocr-torchless

XianScan: github.com/ArbenApura/xianscan-rust

gnurt MangaOCR (~8 MB): github.com/gnurt2041/MangaOCR

manga-ocr-rs: lib.rs/crates/manga-ocr-rs

MIT 48px CTC ONNX: huggingface.co/banned404/mit48pxctc-ocr-onnx

trocr-small-korean: huggingface.co/team-lucid/trocr-small-korean

chineseocr_lite: github.com/ouyanghuiyu/chineseocr_lite

ZentrixLabs.OcrCorrection: github.com/ZentrixLabs/ZentrixLabs.OcrCorrection

Comic bubble detector: huggingface.co/ogkalu/comic-text-and-bubble-detector

ComicTL (browser): comictl.kiuyha.my.id

9. Bottom-Line Summary

1. Detect & crop speech bubbles.
2. Specialized comic OCR: **Baberu** (bubbles JA/ZH/EN), **Hayai LiteRT** (KR+CJK), **RapidOCR** (bulk), or ultra-light gems (gnurt ~8 MB, chineseocr_lite ~4.7 MB, manga-ocr-rs).
3. English residual errors → **ZentrixLabs** or Macula / dictionary rules.
4. Full chapter automation → **XianScan** (CPU-first, manhwa-aware).

All listed tools are open-source or freely usable, run offline on CPU, and are actively used or benchmarked on comic material as of 2026. Start with bubble cropping + Baberu or RapidOCR for the best accuracy/speed/footprint balance; reach for the hidden gems when you need extreme size, pure Rust, or language-specific minimal models.

Expanded report aggregates public benchmarks, GitHub/Hugging Face docs, scanlation toolchains and production manhwa pipelines. Re-benchmark on your own panels — font style and scan quality vary widely.